

Electricity and Magnetism

Equations

$$\boxed{\mathbf{F}_e = k_e \frac{|q_1 q_2|}{r^2} \hat{r}} \quad k_e \triangleq \frac{1}{4\pi\epsilon_0} = 9(10^9) \frac{Nm^2}{C^2}$$

$$\mathbf{F}_e = q\mathbf{E} \quad \mathbf{E}_{\text{net}} = \sum_i k_e \frac{q_i}{r_i^2} \hat{r} = \int k_e \frac{dq}{r^2} \hat{r}$$

$$\lambda = \frac{dq}{dL} \quad \eta = \frac{dq}{dA} \quad \rho = \frac{dq}{dV} \quad \mathbf{E}_{\text{plane}} = \frac{|\eta|}{2\epsilon_0}$$

$$\mathbf{E}_c = \frac{Q}{\epsilon_0 A} \quad \Phi_s \equiv \oint \mathbf{E} \bullet d\mathbf{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$U_e = \sum_{\text{pairs}}^{\text{all}} k_e \frac{q_i q_j}{r_{ij}} \quad V \triangleq \frac{U_{q+\text{sources}}}{q}$$

$$\Delta V = - \int_a^b \mathbf{E} \bullet d\mathbf{s} \quad U = qV \quad \mathbf{E} = -\vec{\nabla}V$$

$$W = q\mathbf{E}_s \Delta s \quad \Delta U = -q\mathbf{E}_s \Delta s \quad \Delta V = -\mathbf{E}_s \Delta s$$

$$C \triangleq \frac{Q}{\Delta V} \quad C_c = \frac{\epsilon_0 A}{d} \quad U_c = \frac{1}{2} C \Delta V^2$$

$$I(t) = -\frac{dQ}{dt} \quad J \triangleq \frac{I}{A} \quad J_{\text{total}} = J_{\text{portion}}$$

$$\boxed{J = \sigma \mathbf{E} \iff V = IR}$$

$$R = \frac{L}{\sigma A} = \frac{\rho L}{A} \quad \rho\sigma = 1$$

$$P \triangleq \frac{dE}{dt} = IV = I^2 R = \frac{V^2}{R}$$

$$\text{Kirchoff: } \boxed{\Sigma I_{\text{in}} = \Sigma I_{\text{out}} \quad \Sigma V_{\text{closed loop}} = 0}$$

$$\text{Biot-Savart Law: } \boxed{\mathbf{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{s} \times \hat{r}}{r^2}}$$

$$\vec{\mu} \triangleq NI\vec{A} \quad \vec{A} = \hat{n}A \quad \boldsymbol{\tau} = \vec{\mu} \times \mathbf{B} \quad \tau_L = \frac{L}{R}$$

$$\mathbf{B}_{\text{solenoid}}(t) = \mu_0 n I(t) \quad n = \frac{N}{l} \quad \mathbf{F}_B = q\mathbf{v}_+ \times \mathbf{B}$$

$$\mathbf{F}_B = I\mathbf{L} \times \mathbf{B} \quad \Phi_B = \int \mathbf{B} \bullet d\vec{A}$$

$$\boxed{\varepsilon_{\text{ind}} = \oint \mathbf{E} \bullet d\vec{s} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt} \int \mathbf{B} \bullet d\vec{A}}$$

$$\varepsilon_L = -L \frac{dI}{dt} \quad \Phi_B = LI \quad P_L = I\varepsilon_L$$

$$\omega = 2\pi f \quad \omega = \sqrt{\frac{1}{LC}} \quad I_{\text{max}} = Q_0 \omega$$

$$Q(t) = Q_0 \cos(\omega t) \quad I(t) = I_{\text{max}} \sin(\omega t)$$

Charging Capacitor

$$q(t) = q_{\infty}(1 - e^{-t/\tau}) \quad I(t) = \frac{dq}{dt} = I_0 e^{-t/\tau}$$

Discharging Capacitor

$$q(t) = q_0(e^{-t/\tau}) \quad I(t) = \frac{dq}{dt} = -I_0 e^{-t/\tau}$$

Series Circuits

$$R_T = R_1 + R_2 + \dots + R_n$$

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

$$Q_1 = Q_2 = \dots = Q_n$$

$$L_T = L_1 + L_2 + \dots + L_n$$

Parallel Circuits

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

$$C_T = C_1 + C_2 + \dots + C_n$$

$$Q_T = Q_1 + Q_2 + \dots + Q_n$$

$$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_n}$$

Using Gauss' Law

1. Find the direction of the electric field
2. Choose a Gaussian Surface that satisfies the following criteria for all points on the surface:
 - The electric field has a constant value
 - The electric field is perpendicular to the surface

This allows the integral to simplify to:

$$\Phi_E = \oint \mathbf{E} \cdot d\vec{A} \implies \mathbf{E} = \frac{q_{in}}{\epsilon_0 A_{G.S.}}$$

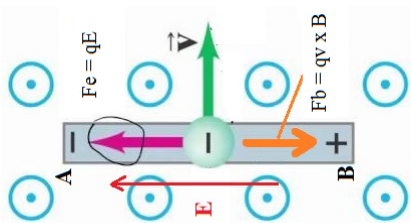
3. Find the total charge inside the Gaussian Surface (q_{in})

Using Ampère's Law

1. Using symmetry, sketch $\mathbf{B}$
2. Draw an Amperian loop passing through point of interest (choose such that it results $\mathbf{B} \cdot d\vec{s}$ or 0)
3. Find net current through the loop
4. Apply Ampère's Law

$$\oint \mathbf{B} \cdot d\vec{s} = \mu_0 I_{net}$$

Motional ϵ MF



1. Motion of bar causes $\mathbf{F}_B$. $\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}$
2. $\mathbf{F}_B$ pushes positives to one end. The negatives go to the other end.
3. This causes a voltage between the two ends. $\Delta V = - \int \mathbf{E} \cdot d\vec{l} = \mathcal{E}$
4. $\mathbf{F}_B = \mathbf{F}_E \implies qvB = qE \implies vB = E$
5. $\Delta V = \mathcal{E} = vBL$

The potential difference across the conductor $\equiv$ motional emf

$$P_{in} = \mathbf{F}v_0 = \left(\frac{v_0 \mathbf{B} L}{R} \right) v_0 \mathbf{B} L = I^2 R = P_{out}$$

Lenz's Law

$$\epsilon_{ind} = - \frac{d\Phi_B}{dt} \quad \text{OR} \quad \mathcal{E} = - \frac{d\Phi_B}{dt}$$

1. Left thumb towards increasing $\mathbf{B}$
2. Right thumb anti-parallel to left thumb
3. Right hand finger curl is direction of I_{ind}

Maxwell's Laws

1. Gauss' Law

$$\Phi_E = \oint \mathbf{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

2. Gauss's Law for Magnetism

$$\oint \mathbf{B} \cdot d\vec{A} = 0$$

3. Faraday's Law

$$\oint \mathbf{E} \cdot d\vec{s} = - \frac{d\Phi_B}{dt}$$

4. Ampère-Maxwell Law

$$\oint \mathbf{B} \cdot d\vec{s} = \mu_0 \left(I + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

Plus bonus:

- Lorentz Force

$$\mathbf{F} = q_+ (\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Waves

$$\mathbf{E}_{xx} = \epsilon_0 \mu_0 \mathbf{E}_{tt} \quad \mathbf{E}(x, t) = \mathbf{E}_0 \cos(kx - \omega t)$$

$$\mathbf{B}_{xx} = \epsilon_0 \mu_0 \mathbf{B}_{tt} \quad \mathbf{B}(x, t) = \mathbf{B}_0 \cos(kx - \omega t)$$

$$\mathbf{E}_{max} = \mathbf{B}_{max} c \quad \omega = 2\pi f \quad k \triangleq \frac{2\pi}{\lambda}$$

$$v = \lambda f \quad v = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \triangleq c \quad 1 = fT \quad f = \frac{E}{h}$$

$$U_E = \frac{1}{2} \epsilon_0 \mathbf{E}_0^2 \quad U_B = \frac{1}{2 \mu_0} \mathbf{B}_0^2 \quad U_T = \epsilon_0 \mathbf{E}^2$$

$$\mathbf{S} \triangleq \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \quad |\mathbf{S}| = \frac{1}{A} \frac{dU}{dt} \quad \mathbf{S}_{\max} = \frac{\mathbf{E}_0^2}{c \mu_0}$$

$$\mathbf{S}_{\text{avg}} = I = \frac{P}{A} = \frac{\mathbf{E}_0^2}{2c\mu_0} = \frac{c\epsilon_0 \mathbf{E}_0^2}{2}$$

$$I = U_{\text{avg}} c \quad I_{\text{point source}} = \frac{P_{\text{source}}}{4\pi r^2}$$

Radiation Pressure

$$E = mc^2 \implies m = \frac{E}{c^2}$$

$$p = mv \implies p = \frac{E}{c^2} c = \frac{E}{c}$$

$$\mathbf{F} = \frac{dp}{dt} = \frac{1}{c} \frac{dE}{dt} = \frac{1}{c} IA$$

$$\mathbf{F} = \frac{1}{c} IA \quad \sigma = \frac{I}{c}$$

Additional Notes

- Electric field lines point in the direction of decreasing electrical potential
- After a long time:
 - Capacitors behave like batteries
 - Inductors/coils behave like wires
 - Resistors
- Antiparallel currents repel, parallel currents attract